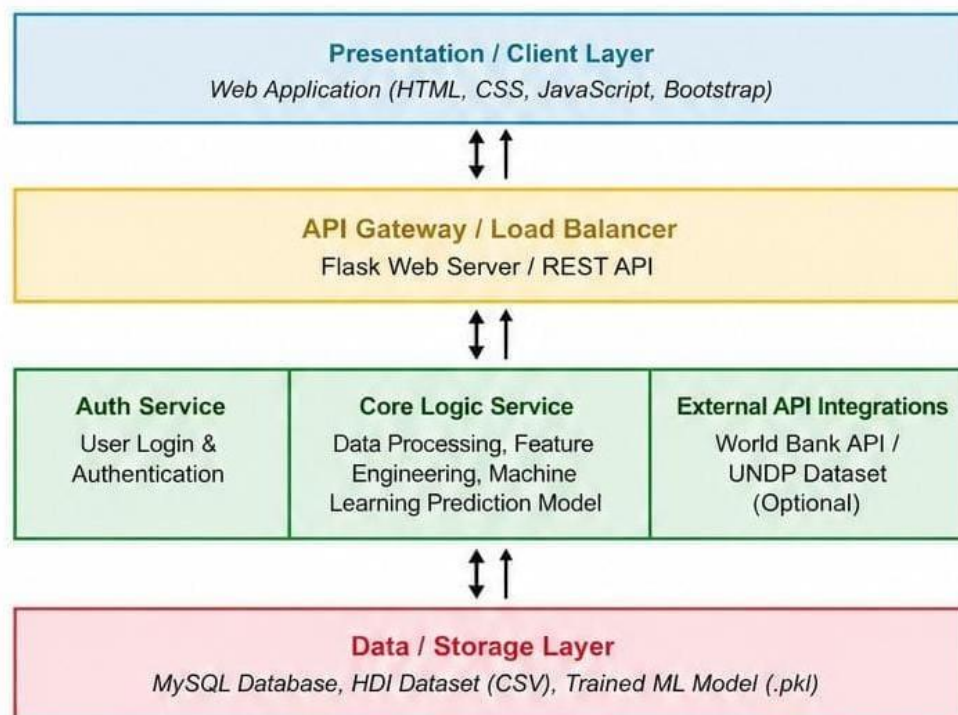


Solution Architecture

Date	3 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	5 Marks

Solution Architecture Diagram:

The solution architecture of the HDI Prediction System consists of a presentation layer, application layer, machine learning services, and a data storage layer. User requests are processed through the application server, where the machine learning model predicts the Human Development Index based on socio-economic indicators. The prediction results are then stored and displayed through an interactive web interface..



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering data and viewing HDI predictions.	HTML, CSS, JavaScript, Bootstrap
API Gateway	Handles user requests and connects the frontend with the machine learning backend.	Flask REST API
Microservices	Processes input data, performs prediction, and returns HDI results.	Python, Scikit-learn, Pandas, NumPy
Database	Stores datasets, prediction history, trained models, and user information.	MySQL, CSV Dataset